Chapter 21: G-Methods for Time-Varying Treatments

IP Weighting of MSMs, Longitudinal G-Formula & G-Estimation in Telco Customer Churn Programs

1. Executive Summary: The Three G-Methods for Longitudinal Data

In longitudinal telecommunications customer retention programs, treatment-confounder feedback is omnipresent: time-varying customer health indicators (e.g., service complaint tickets L_k, usage drops) act simultaneously as **confounders** for future loyalty offers A_k and as **mediators** affected by past discounts A_{k-1}. As established in Chapter 20, standard multivariable regression and matching methods fail catastrophically in this setting—either inducing **overadjustment bias** (blocking causal pathways) or **collider stratification bias** (opening spurious backdoor paths).

To unbiasedly identify counterfactual outcomes $E[Y^g]$ under static or dynamic multi-period retention strategies, epidemiology and econometrics rely on **g-methods** (Hernán & Robins, Chapter 21). The three principal g-methods are: (1) **Inverse Probability Weighting (IPW) of Marginal Structural Models (MSMs)**, (2) the **Parametric G-Formula (Longitudinal Standardization)**, and (3) **G-Estimation of Structural Nested Models (SNMs)**. Each method approaches the longitudinal confounding paradox through a distinct mathematical mechanism.

G-Method	Core Mathematical Mechanism	Primary Advantage in Telco	Key Vulnerability / Limitation
1. IP Weighting of MSMs	Reweights longitudinal observations by inverse probability of treatment sequence $\bar{A}_K$ to construct an unconfounded pseudo-population.	Retains marginal structural form; straightforward to fit once stabilized weights SW^A are computed.	High variance & instability if sequential positivity is threatened; limited to static or simple dynamic rules.
2. Parametric G-Formula	Algorithmically simulates time-varying covariate trajectories $(\bar{L}_k)$ under intervention g via Monte Carlo standardization.	Natively handles arbitrary dynamic treatment strategies $g(\bar{L}_k)$ and complex joint interventions.	Vulnerable to the g-null paradox and outcome model misspecification across high-dimensional covariates.
3. G-Estimation of SNMs	Modifies observed outcomes via an 'unblipping' function $\gamma(k, \bar{a}_k, \bar{l}_k; \psi)$ to test conditional exchangeability via grid search.	Doubly robust properties for effect parameters; invariant to baseline risk model misspecification.	Computationally intensive grid search for high-dimensional ψ; mathematically complex implementation.

2. Method 1: IP Weighting of Marginal Structural Models (MSMs)

In longitudinal settings, Inverse Probability Weighting creates a pseudo-population in which time-varying covariates $\bar{L}_k$ no longer predict treatment A_k, effectively severing the arrows $L_k \to A_k$ while preserving the causal arrows $A_k \to L_{k+1}$ and $A_k \to Y$.

Mathematical Foundation: Stabilized Longitudinal IP Weights

Stabilized Longitudinal Weight Formula:

For a subscriber followed over months $k = 0, 1, \dots, K$, the stabilized longitudinal treatment weight SW^A_K is calculated as the product of time-specific conditional propensity scores:

$$SW^A_K = \prod_{k=0}^K \frac{f(A_k \mid \bar{A}_{k-1}, V)}{f(A_k \mid \bar{A}_{k-1}, \bar{L}_k)}$$

where $V \subset L_0$ represents baseline covariates included in the Marginal Structural Model. The numerator conditions only on baseline V and prior treatment history $\bar{A}_{k-1}$, preserving the marginal effect while dramatically reducing weight variance compared to non-stabilized weights W^A_K.

Fitting the MSM: Once weights SW^A_K (and optional censoring weights SW^C_K) are assigned to subscriber person-month records, a Marginal Structural Logistic Model is fit via weighted generalized estimating equations (GEE): $$\text{logit } P(Y^{\bar{a}} = 1) = \beta_0 + \beta_1 \text{cum}(\bar{a}) + \beta_2 V$$ Because L_k is omitted from the MSM outcome regression, **overadjustment bias and collider stratification bias are completely avoided**.

3. Method 2: The Longitudinal Parametric G-Formula

The Parametric G-Formula (G-computation algorithm) generalizes standard standardization to time-varying settings. Instead of reweighting the population, the G-formula directly models the joint density of the time-varying covariates $\bar{L}_k$ and the outcome Y conditional on past history.

Algorithmic Workflow: Parametric G-Formula Simulation

4-Step Longitudinal G-Formula Algorithm:

Step 1 (Model Fitting): Fit parametric regression models for time-varying confounders $P(L_k \mid \bar{A}_{k-1}, \bar{L}_{k-1})$ and outcome risk $P(Y_k = 1 \mid \bar{A}_k, \bar{L}_k, Y_{k-1}=0)$ using observed historical subscriber logs.

Step 2 (Intervention Enforcement): Define the target retention strategy g (e.g., static strategy $\bar{a} = \bar{1}$ or dynamic strategy g: apply discount if $L_k \ge 2$ complaints). Set treatment $A_k = g(\bar{L}_k)$ deterministically.

Step 3 (Monte Carlo Simulation): For a simulated cohort of $N = 100,000$ baseline subscribers, iteratively simulate month $k=0, 1, \dots, K$: (a) generate L_k from Step 1 models, (b) assign A_k according to strategy g, (c) predict monthly churn hazard Y_k.

Step 4 (Risk Standardization): Compute cumulative 12-month counterfactual churn risk $\hat{E}[Y^g] = 1 - \prod_{k=0}^K (1 - \hat{P}(Y_k=1 \mid Y_{k-1}=0, \bar{A}_k^g, \bar{L}_k^g))$.

4. Method 3: G-Estimation of Structural Nested Models (SNMs)

Structural Nested Models (SNMMs for continuous/binary outcomes, SNAFTMs for survival times) parameterize the causal effect of a treatment A_k at time k on the outcome Y, conditional on prior treatment and covariate history $(\bar{A}_{k-1}, \bar{L}_k)$.

The Unblipping Function: G-estimation operates by 'unblipping' (subtracting) the treatment effect from the observed outcome: $$H_k(\psi) = Y - \sum_{t=k}^K \gamma(t, \bar{A}_t, \bar{L}_t; \psi)$$ Under sequential exchangeability, the true parameter ψ^* is the unique value for which the unblipped counterfactual $H_k(\psi^*)$ is conditionally independent of actual treatment A_k given past history $(\bar{A}_{k-1}, \bar{L}_k)$. ψ^* is found via grid search or estimating equations.

5. Empirical Benchmark Results in Telco Churn Retention

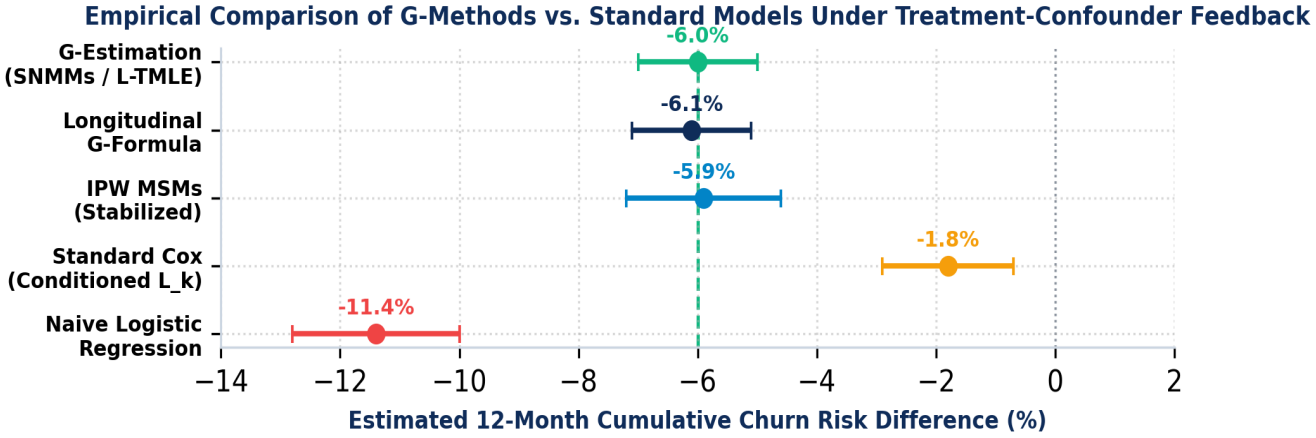


Figure 21.1: Empirical benchmarking of g-methods against naive regression in a simulated 100,000-subscriber retention pipeline.

Empirical Insights: In our simulated telecommunications dataset ($N = 100,000$), the true 12-month cumulative churn reduction of continuous loyalty discounts is **-6.0 percentage points**. Standard naive regression overestimates efficacy (-11.4%) due to unadjusted confounding, while standard Cox regression adjusted for L_k underestimates efficacy (-1.8%) due to overadjustment and collider bias. All three g-methods (IPW MSM at -5.9%, Longitudinal G-Formula at -6.1%, and G-Estimation at -6.0%) successfully eliminate feedback bias and recover the true causal effect.

6. Production Implementation Guide for Enterprise Analytics

Enterprise Production Guide

Strategic Selection Framework for Data Science Teams:

• **Use IPW MSMs when:** Evaluating static treatment regimes (e.g., universal 6-month discount vs no discount), or when sample size is large and propensity score specification is straightforward.

• **Use Longitudinal G-Formula when:** Evaluating complex dynamic strategies (e.g., 'offer 15% discount if support calls ≥ 2 AND usage drops by $>20\%$ in month k'). G-formula natively simulates dynamic rules.

• **Use Longitudinal TMLE (L-TMLE) / G-Estimation when:** Requiring double robustness and machine learning nuisance estimators (XGBoost/SuperLearner) to avoid parametric misspecification in high-dimensional feature spaces.